

Quiz 1 – Kinematics

Problem 1 (25 pts)

Plot the velocity of the object, whose acceleration is shown in Figure on the right. The velocity is zero at the moment $t = 0$ s. Find the mean velocity averaged over the time interval $[6, 14]$ s.

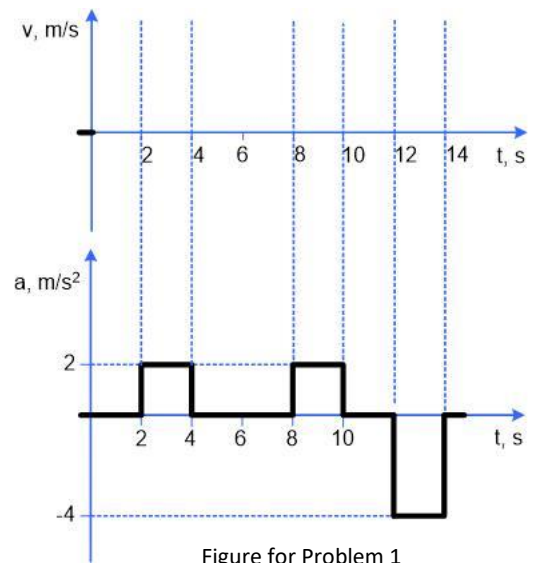


Figure for Problem 1

Problem 2 (25 pts)

The tram starts moving with a constant acceleration of 0.4 m/s^2 . How long can a passenger stay after the beginning of the tram movement, then catch up to him, running at a constant speed of 10 m/s ?

Problem 3 (25 pts)

A drop of rain falls at an angle of $\alpha_1 = 30^\circ$ to the vertical axis when the wind speed is $u_1 = 10 \text{ m/s}$. What is the angle α_2 at which the drop falls after the wind has been increased to $u_2 = 20 \text{ m/s}$? Assume that the velocity of the raindrops with respect to the air is constant.

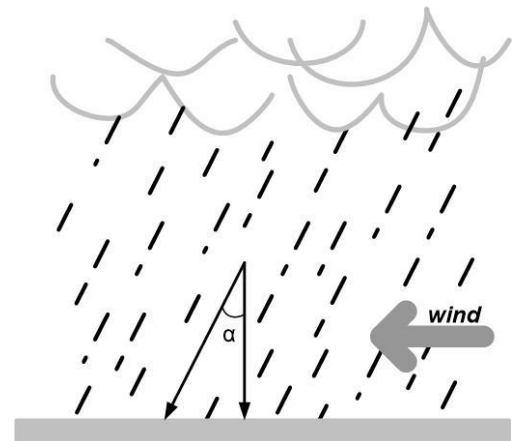


Figure for Problem 3

Problem 4 (25 pts)

A particle moves in uniform circular motion over a horizontal xy plane. It takes 20 s to make a full circle. At one instant, it moves through the point with coordinates $[4 \text{ m}, 4 \text{ m}]$ with the acceleration of $(0 \hat{i} + 8 \hat{j}) \text{ m/s}^2$. Find the x and y coordinates of the center of the circular path.